Yixiao Wang

CV

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EDUCATION

Duke University, Graduate School — Aug. 2024 - May. 2026 (Expected)
M.S., Statistical Science — GPA: 4.0/4.0, Ranking: 1/45

University of Science and Technology of China, School of the Gifted Young — Sep. 2020 - Jul. 2024
B.S., Mathematics and Applied Mathematics **(Outstanding Graduate)**
Major in Probability and Statistics track — GPA (WES Converted): 3.91/4.0, Ranking: 9/92

RESEARCH INTERESTS

Statistical and Machine Learning Theory, **Functional Data Analysis.**

RESEARCH

Enhanced Cyclic Coordinate Descent | Co-First Author (In Progress) — Dec. 2024 - Present
Supervised by Assistant Professor Aditya Devarakonda at Wake Forest University.

- Proposed and developed a novel Taylor expansion-based approximation method to replace block coordinate descent (BCD) in GLM with elastic net/LASSO, ensuring convergence while accelerating computation, achieving a 2–3× speedup over GLMnet in R to date.
- Fully responsible for algorithmic development and theoretical proofs.
- Aiming to complete this work by April 2025.

Stability-guided Adaptive Diffusion Acceleration | Co-First Author (ICML 2025 Submission) — Mon. 2025
Collaborated with students in Professor Yiran Chen's group at Duke University.

- Developed a stable guided method based on probability flow ODE to accelerate diffusion models by selecting appropriate skipping strategies and providing approximations in the sampling process.
- Ensured stability across various solvers, including DPM-Solver++ and the Euler solver.s
- Fully responsible for algorithmic development and theoretical proofs.

Trimmed Mean for Partially Observed Functional Data | Bachelor's Thesis — Feb. 2024 - May. 2024
Supervised by Associate Professor Xiaohong Lan from University of Science and Technology of China.

- Defined the Trimmed Mean for partially observed functional data.
- Proved strong consistency of depth function for partially observed functional data and the trimmed mean.
- Conducted numerical simulations experiment and corresponding R code.
- Thesis: arXiv | Code: GitHub.

RELEVANT COURSEWORK

Mathematics & Statistics: Probability Theory (Original & Advanced, Honors, both A+), Advanced Probability Theory (A+), Mathematical Statistics (Original & Advanced, Honors, both A+), Bayesian Analysis, Time Series Analysis (A), Stochastic Processes (A-), Predictive Inference (A), Statistical Inference (A), Functional Analysis (A-), Real Analysis (A-), Complex Analysis, Mathematical Analysis I (A+), II (A), III (A-), Abstract Algebra (A-), Linear Algebra I (A+), II (A-), Fundamentals of Algebra (A), Differential Equations (A+), Operations Research, Differential Geometry (A), Fundamentals of Geometry (A).
Computer Science: Machine Learning Theory & Algorithms (A), Data Structures and Databases (A), C Programming (A+), R Programming (A), Python Programming (A), MATLAB (A-), Linux (A-).

PROJECTS

Airbnb Price Prediction in New York City| Course Project for *Machine Learning* — Nov. 2024

- Report: PDF | Code: GitHub.

Basketball Breakdown: NCAA Shiny App| Course Project for *R Language* — Dec. 2024

- App: NCAA Data Archive | Report: PDF | Code: GitHub.

LSTM-GRU Hybrid Network | Course Project for *Time Series Analysis* — Jun. 2024

- Report: PDF | Code: GitHub.

TEACHING & PRESENTATION

Linear Algebra B1 | Teaching Assistant Mar. 2023 - Jul. 2023

- Excellent Teaching Assistant (Spring 2023): Ranked in the top 3 at USTC, among over 1000 TAs.

Probability Theory Seminar | Presenter April. 2023

- Slides: pdf.

AWARDS & HONORS

Outstanding Graduate from the University of Science and Technology of China (USTC), Class of 2024 **(Top 20%).**
China Petroleum Scholarship (2023): Awarded to **only 3** students in the School of the Gifted Young at USTC.
Excellent Teaching Assistant (Spring 2023): Ranked in the **top 3** at USTC, among over 1000 TAs.
Silver Prize for Outstanding Student Scholarship (2022 and 2021): Awarded to the **top 15%** of students at USTC.
First Prize in the Chinese Mathematical Competitions for University Students (2022 and 2021): Ranked in the **top 1%** of all participants.
Excellent President of the School Club (2022).
QiangWeiFengGongDeYu (Diligence and Moral Conduct) Scholarship(2022).
Bronze Prize for Outstanding Freshmen Scholarship (2020): Awarded to the **top 35%** of students at USTC.

ACTIVITIES & LEADERSHIP

Origami Club, USTC | President Jun. 2021- Jun. 2022

- Excellent President of the School Club (2022).

Hefei Chunyu Parent Support Center for Intellectually Disabled Children | Volunteer Sep. 2021- Jan. 2022

- Played basketball with autistic children and helped coaches to keep order in class once per week.

ADDITIONAL INFORMATION

Language: Chinese (Mandarin and Sichuanese dialect), English (TOEFL 103 with 23 in Speaking).
Programming & Software Skills: Python (mainly), C, MATLAB, R, Latex, Linux, HTML, CSS, SPSS, Office, etc.
Interests: Origami, click here to see my artwork.